

Issue # 7: Numerator Data - Quality Check and Analysis by Diabetes Status 2023-2024

Ron Huang

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This Write-up contains tables and plots for quality check of Numerator data and descriptive analysis of expenditure by diabetes status. The original user-trip-item level data is transformed to user- month level data, with variables summarized at the monthly level for each user. SNAP and Non-SNAP treatment is determined by SNAP participation rate of each user over the periods of the data (>0.5 on SNAP; ≤ 0.5 Non-SNAP); Not adjusted for inflation

1. Figures: Quality Check

- Monthly attrition
- Attrition between cohorts
- Monthly net change
- Counts of dropouts and addition between cohorts
- EBT spending correlation (bin scatter)
- Ration of food spending to EBT spending (distribution of inframarginal SNAP household)

2. Table: Quality Check

- 2023 Annual Expenditure in comparison to CEX

3. Figures: Trend analysis of expenditure by diabetes status

- level/log total spending winsorized
- level/log Food spending winsorized
- level/log Food on EBT winsorized
- level/log Soda spending winsorized
- level/log Soda on EBT winsorized
- prop. food spending
- prop. food on EBT
- prop. soda
- prop. soda on EBT

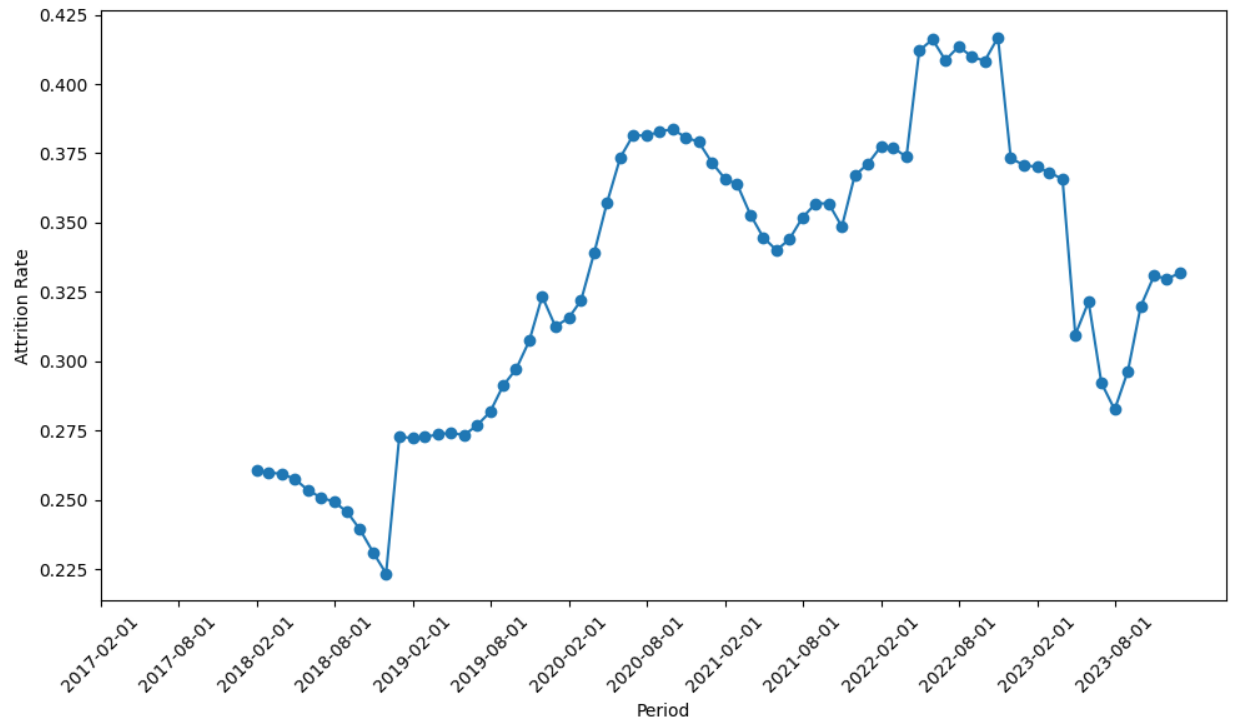
- level/log soda spending by SNAP/Non-SNAP
- level/log soda spending by state $\times$ SNAP

4. Tables: Descriptive analysis of expenditure by diabetes status

- Demographics by treatment status
- Spending by treatment status
- Soda spending by treatment status

Figure 1: Attrition

(a) Between Cohort



(b) Monthly

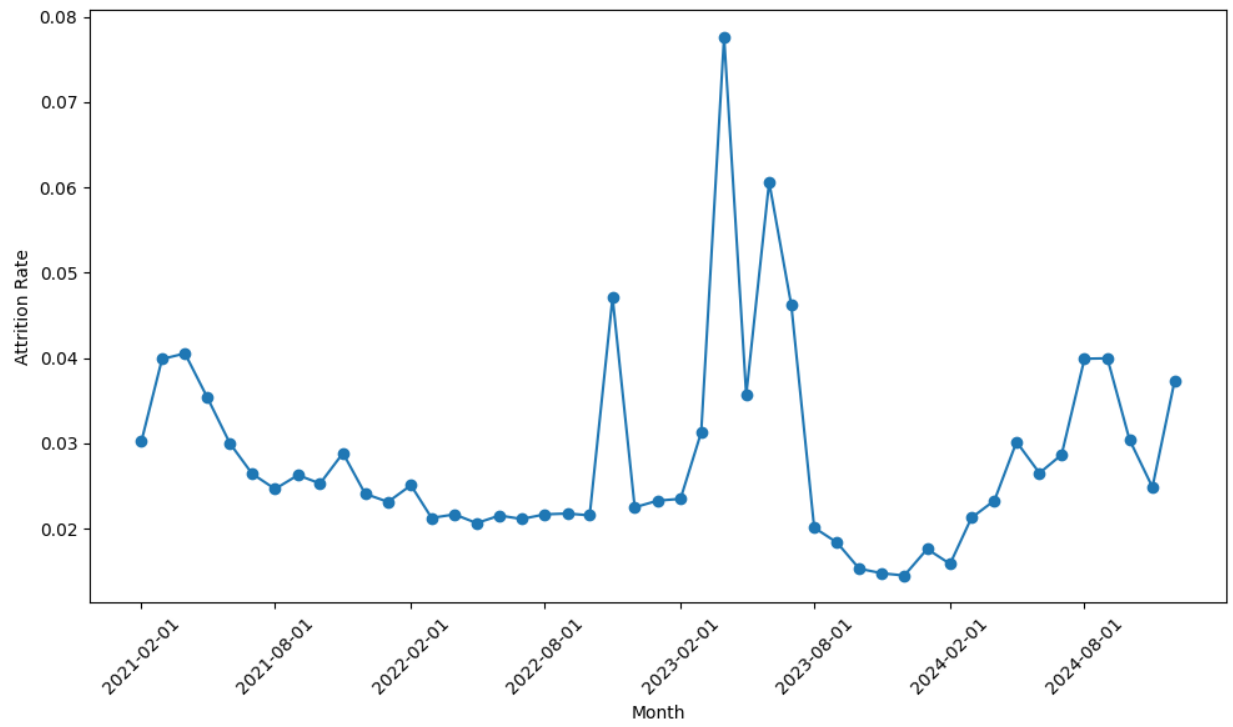
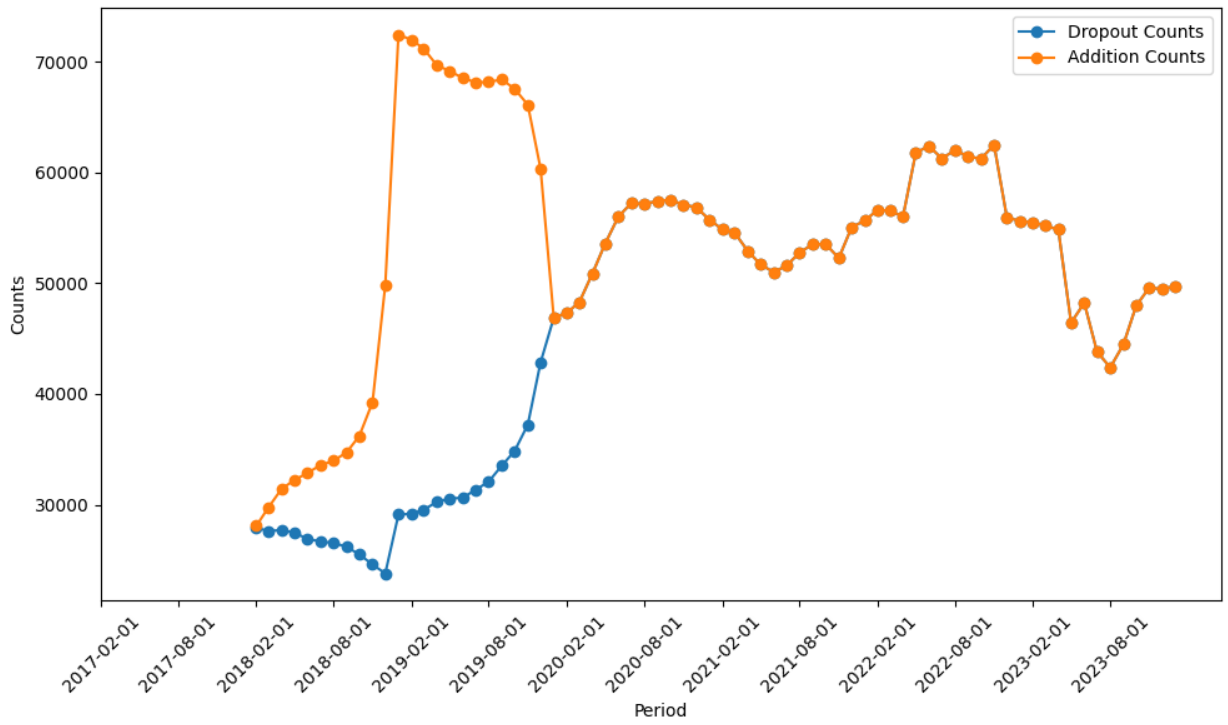


Figure 2: Net Changes

(a) Level



(b) Rate

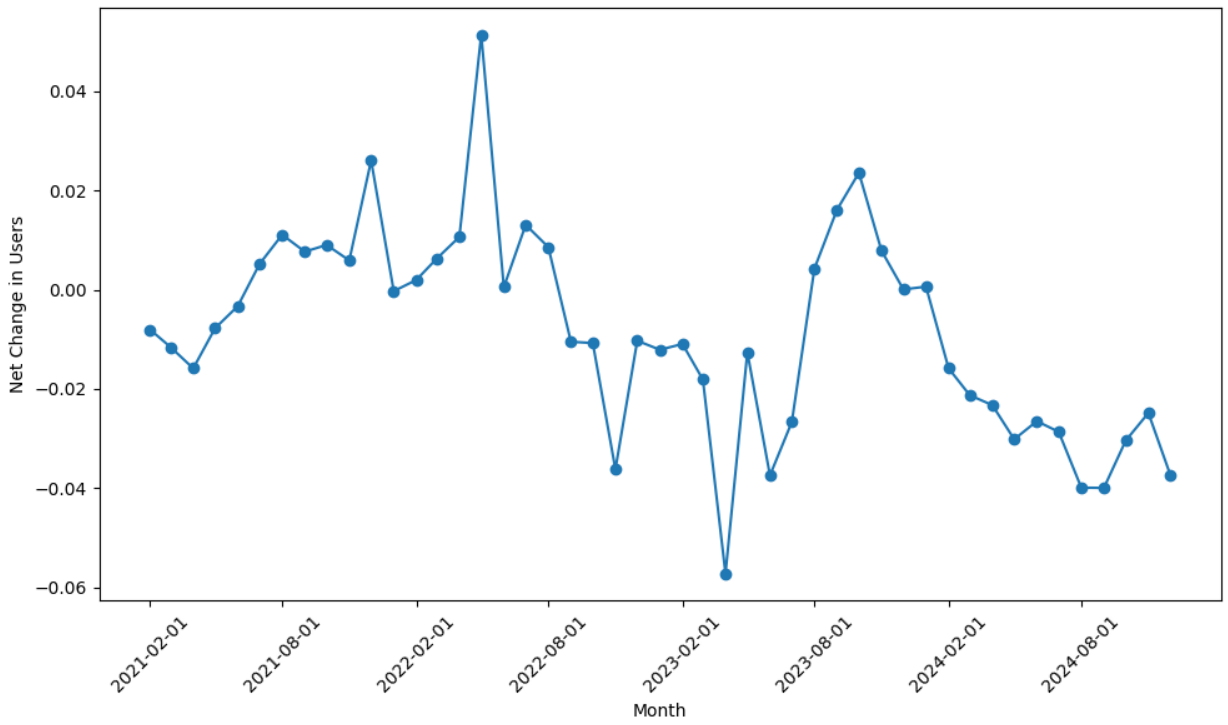
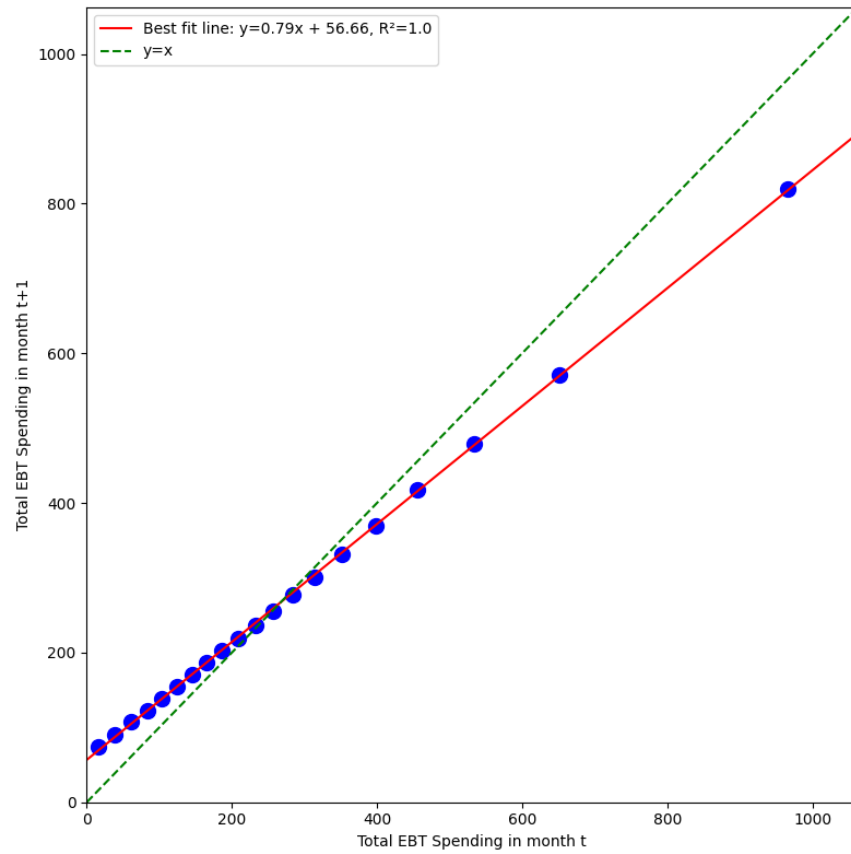


Figure 3: EBT Spending

(a) Monthly Correlation



(b) Proportion of Food Spending on EBT

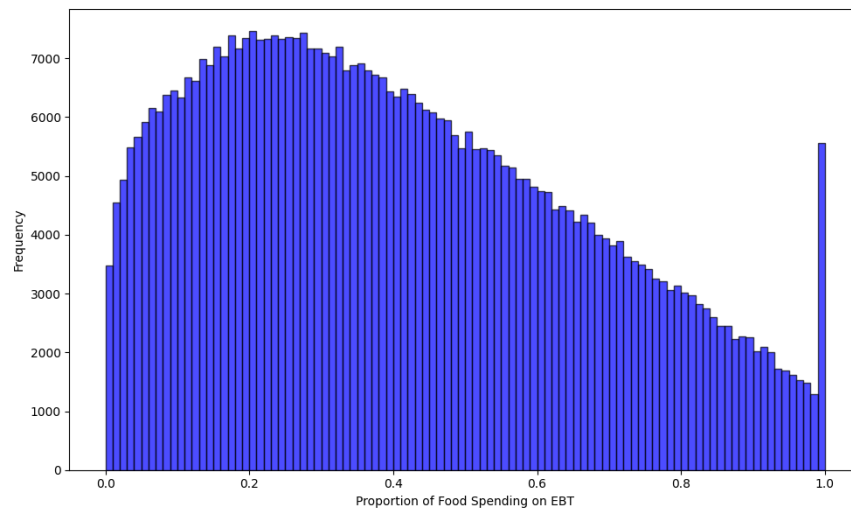


Table 1: Annual Expenditure in 2023

	Total	<15k	15–29k	30–39k	40–49k	50–69k	70–99k	100–149k	150–199k	200k+
Total Spending	16,956.61 (22,970.10)	14,921.84 (21,658.75)	14,130.09 (19,978.39)	13,907.81 (18,461.29)	14,160.82 (18,606.28)	14,439.09 (19,847.39)	15,480.91 (20,111.03)	18,734.63 (26,562.45)	24,901.28 (27,652.42)	25,960.63 (29,257.26)
Food Spending	6,758.58 (9,108.25)	6,297.14 (9,181.03)	6,007.39 (9,336.73)	5,764.50 (7,939.10)	5,830.90 (7,650.04)	5,859.87 (9,246.84)	6,123.58 (8,033.65)	7,255.13 (9,333.09)	9,533.27 (10,102.21)	9,906.80 (10,942.92)
Food at Home	6,144.95 (8,486.20)	5,674.39 (8,486.89)	5,428.77 (8,824.30)	5,193.74 (7,346.39)	5,250.15 (7,003.62)	5,291.89 (8,698.08)	5,556.99 (7,397.79)	6,637.01 (8,736.01)	8,756.87 (9,443.60)	9,132.91 (10,189.02)
Food Away from Home	613.63 (1,207.37)	622.76 (1,244.22)	578.63 (1,173.66)	570.76 (1,084.51)	580.75 (1,166.10)	567.98 (1,162.19)	566.59 (1,125.24)	618.12 (1,153.94)	776.41 (1,318.74)	773.89 (1,618.36)
Soda Spending	243.67 (609.29)	263.68 (587.44)	258.81 (512.47)	239.28 (448.09)	226.73 (447.16)	223.95 (1,033.27)	218.92 (439.49)	242.78 (519.88)	295.68 (491.08)	287.38 (545.79)
Share of Food	0.42 (0.13)	0.44 (0.14)	0.44 (0.13)	0.43 (0.13)	0.43 (0.13)	0.42 (0.13)	0.41 (0.13)	0.41 (0.13)	0.41 (0.13)	0.42 (0.14)
Prop. of Food on FAH	0.38 (0.13)	0.40 (0.14)	0.40 (0.13)	0.39 (0.13)	0.39 (0.13)	0.38 (0.13)	0.38 (0.13)	0.38 (0.13)	0.38 (0.14)	0.39 (0.14)
Prop. of Food on FAFH	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.04 (0.04)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)
Prop. of Food on Soda	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)

Notes: This table shows the average annual spending and proportions in 2023.

Figure 4: CEX Annual Expenditure

(a) Monthly Correlation

Table 1203. Income before taxes: Annual expenditure means, shares, standard errors, and relative standard errors, Consumer Expenditure Surveys, 2023											
Item	All consumer units	Less than \$15,000	\$15,000 to \$29,999	\$30,000 to \$39,999	\$40,000 to \$49,999	\$50,000 to \$69,999	\$70,000 to \$99,999	\$100,000 to \$149,999	\$150,000 to \$199,999	\$200,000 and more	
Average annual expenditures											
Mean	\$77,280	\$32,081	\$35,587	\$47,027	\$50,204	\$59,461	\$71,899	\$90,677	\$117,818	\$174,993	
SE	994.16	1,385.24	1,054.11	1,212.88	1,058.24	1,163.64	1,243.75	1,218.89	2,832.90	3,968.96	
RSE	1.29	4.32	2.96	2.58	2.11	1.96	1.73	1.34	2.40	2.27	
Food											
Mean	9,985	5,684	5,181	6,665	7,302	8,413	9,705	11,960	15,264	18,525	
Share	12.9	17.7	14.6	14.2	14.5	14.1	13.5	13.2	13.0	10.6	
SE	122.90	328.22	233.61	323.48	370.02	222.37	308.20	207.67	464.66	493.89	
RSE	1.23	5.77	4.51	4.85	5.07	2.64	3.18	1.74	3.04	2.67	
Food at home											
Mean	6,053	4,037	3,624	4,430	4,769	5,536	6,080	7,053	8,622	9,702	
Share	7.8	12.6	10.2	9.4	9.5	9.3	8.5	7.8	7.3	5.5	
SE	84.91	199.09	185.00	286.97	258.02	181.16	229.41	168.09	258.50	291.30	
RSE	1.40	4.93	5.10	6.48	5.41	3.27	3.77	2.38	3.00	3.00	

(b) CEX Annual Expenditure - FAFH and Beverage

Table 1203. Income before taxes: Annual expenditure means, shares, standard errors, and relative standard errors, Consumer Expenditure Surveys, 2023											
Item	All consumer units	Less than \$15,000	\$15,000 to \$29,999	\$30,000 to \$39,999	\$40,000 to \$49,999	\$50,000 to \$69,999	\$70,000 to \$99,999	\$100,000 to \$149,999	\$150,000 to \$199,999	\$200,000 and more	
Nonalcoholic beverages											
Mean	663	490	434	514	572	620	677	770	856	976	
Share	.9	1.5	1.2	1.1	1.1	1.0	.9	.8	.7	.6	
SE	12.38	36.54	33.79	38.74	43.37	24.11	35.06	30.93	39.55	49.02	
RSE	1.87	7.45	7.78	7.53	7.59	3.89	5.18	4.02	4.62	5.02	
Food prepared by consumer unit on out of town trips											
Mean	96	31	31	53	57	64	72	112	162	265	
Share	.1	.1	.1	.1	.1	.1	.1	.1	.1	.2	
SE	4.96	4.85	6.80	11.24	12.77	6.17	7.15	10.78	15.04	22.98	
RSE	5.17	15.64	21.63	21.05	22.58	9.58	9.93	9.62	9.30	8.68	
Food away from home											
Mean	3,933	1,647	1,556	2,235	2,533	2,877	3,624	4,907	6,643	8,824	
Share	5.1	5.1	4.4	4.8	5.0	4.8	5.0	5.4	5.6	5.0	
SE	69.03	189.40	98.23	135.78	185.86	124.74	123.17	166.79	331.25	299.42	
RSE	1.76	11.50	6.31	6.07	7.34	4.34	3.40	3.40	4.99	3.39	
Alcoholic beverages											
Mean	637	229	203	346	330	384	512	838	1,269	1,567	
Share	.8	.7	.6	.7	.7	.6	.7	.9	1.1	.9	

Table 3: User-Month Panel - Demographics

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Income	49248.50 (48007.31)	49866.71 (49790.04)	88673.89 (58544.06)	95476.81 (64342.62)	86550.98 (62938.52)	10000.00	275000.00
Household Size	2.96 (1.65)	2.96 (1.68)	2.72 (1.36)	2.78 (1.42)	2.79 (1.45)	1.00	7.00
Age	52.80 (12.33)	48.28 (13.13)	54.12 (12.25)	49.37 (12.94)	50.59 (12.94)	19.00	67.50
Married	0.35 (0.48)	0.31 (0.46)	0.64 (0.48)	0.60 (0.49)	0.57 (0.50)	0.00	1.00
Education (years)	13.40 (1.40)	13.35 (1.44)	13.90 (1.30)	13.96 (1.35)	13.85 (1.37)	10.00	16.00
White	0.49 (0.50)	0.52 (0.50)	0.70 (0.46)	0.73 (0.44)	0.69 (0.46)	0.00	1.00
Black	0.22 (0.41)	0.20 (0.40)	0.10 (0.30)	0.08 (0.28)	0.11 (0.31)	0.00	1.00
Asian	0.05 (0.21)	0.05 (0.23)	0.06 (0.24)	0.07 (0.25)	0.06 (0.24)	0.00	1.00
Hispanic	0.22 (0.41)	0.20 (0.40)	0.12 (0.32)	0.10 (0.30)	0.12 (0.33)	0.00	1.00
Ever Child	0.28 (0.45)	0.31 (0.46)	0.20 (0.40)	0.25 (0.43)	0.25 (0.43)	0.00	1.00
Number of States	52	56	53	56	57		
Number of Users	11,037	48,414	31,465	175,490	266,406		
Average Number of Months per User	18.32	19.00	17.05	17.86	17.99		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 4: User-Month Panel - Spending

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Food Spending	718.31 (514.87)	675.50 (507.37)	619.19 (424.46)	585.52 (426.99)	610.31 (442.36)	0.00	50429.40
Total Spending	1595.87 (1161.51)	1507.97 (1153.81)	1565.93 (1110.22)	1485.95 (1128.68)	1513.52 (1129.46)	0.00	135684.55
Prop. of Total Spending on Food	0.48 (0.17)	0.48 (0.18)	0.43 (0.17)	0.43 (0.18)	0.44 (0.18)	0.00	1.00
Prop. of Food Spending on FAH	0.89 (0.12)	0.90 (0.13)	0.88 (0.14)	0.90 (0.13)	0.90 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.11 (0.12)	0.10 (0.13)	0.12 (0.14)	0.10 (0.13)	0.10 (0.13)	0.00	1.00
Prop of Food on EBT	0.33 (0.25)	0.36 (0.27)	0.16 (0.18)	0.18 (0.20)	0.30 (0.26)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.03)	1.00 (0.03)	1.00 (0.04)	1.00 (0.04)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.03)	0.00 (0.03)	0.00 (0.04)	0.00 (0.04)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.82 (0.39)	0.81 (0.39)	0.07 (0.25)	0.05 (0.23)	0.18 (0.38)	0.00	1.00
Number of States	52	56	53	56	57		
Number of Users	11,037	48,414	31,465	175,490	266,406		

Notes: Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization. 18% of SNAP users indicate to have diabetes. 16% of all users indicate to have diabetes.

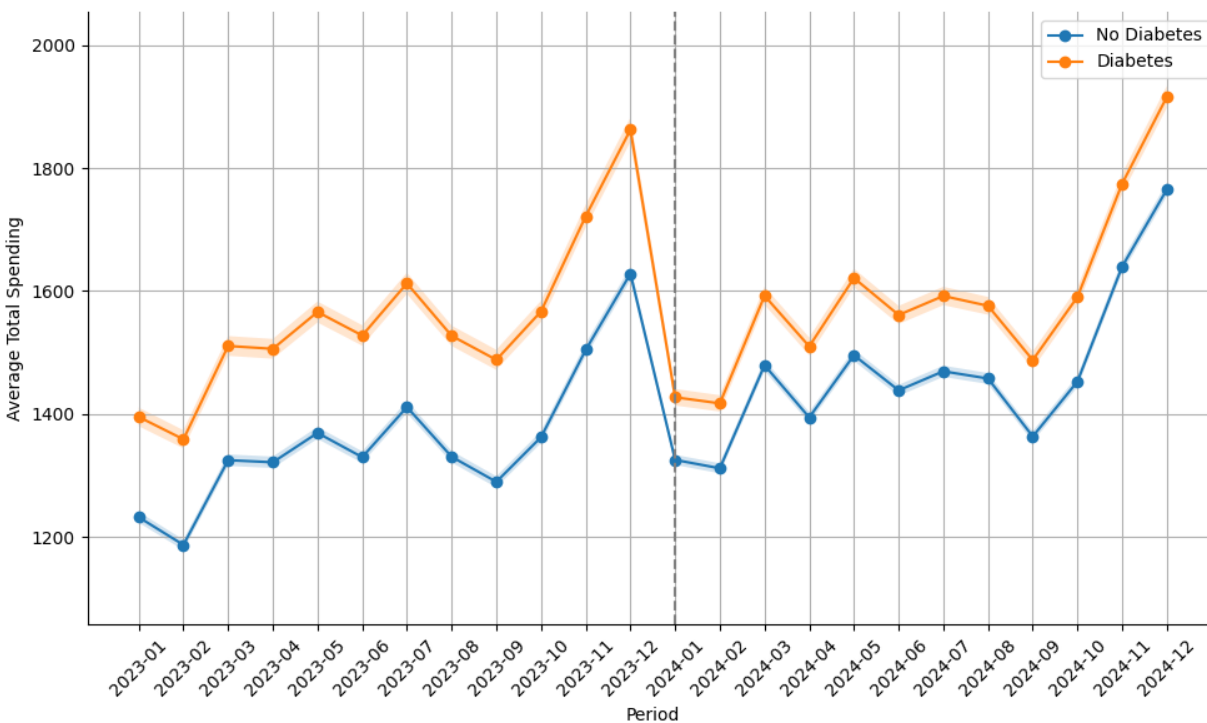
Table 5: User-Month Panel - Soda Spending

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Soda Spending	34.25 (43.39)	29.43 (39.30)	26.61 (39.34)	20.53 (32.63)	23.67 (35.96)	0.00	5749.22
Prop. of Food Spending on Soda	0.05 (0.06)	0.05 (0.06)	0.04 (0.06)	0.04 (0.05)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.89 (0.23)	0.90 (0.23)	0.86 (0.27)	0.87 (0.27)	0.87 (0.27)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.11 (0.23)	0.10 (0.23)	0.14 (0.27)	0.13 (0.27)	0.13 (0.27)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.11)	0.05 (0.10)	0.06 (0.15)	0.06 (0.15)	0.05 (0.12)	0.00	1.00
Prop of Soda Spending on EBT	0.30 (0.34)	0.33 (0.36)	0.15 (0.27)	0.16 (0.29)	0.28 (0.35)	0.00	1.00
Number of Users	11,037	48,414	31,465	175,490	266,406		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Figure 5: Monthly Average of Total Spending

(a) level



(b) log

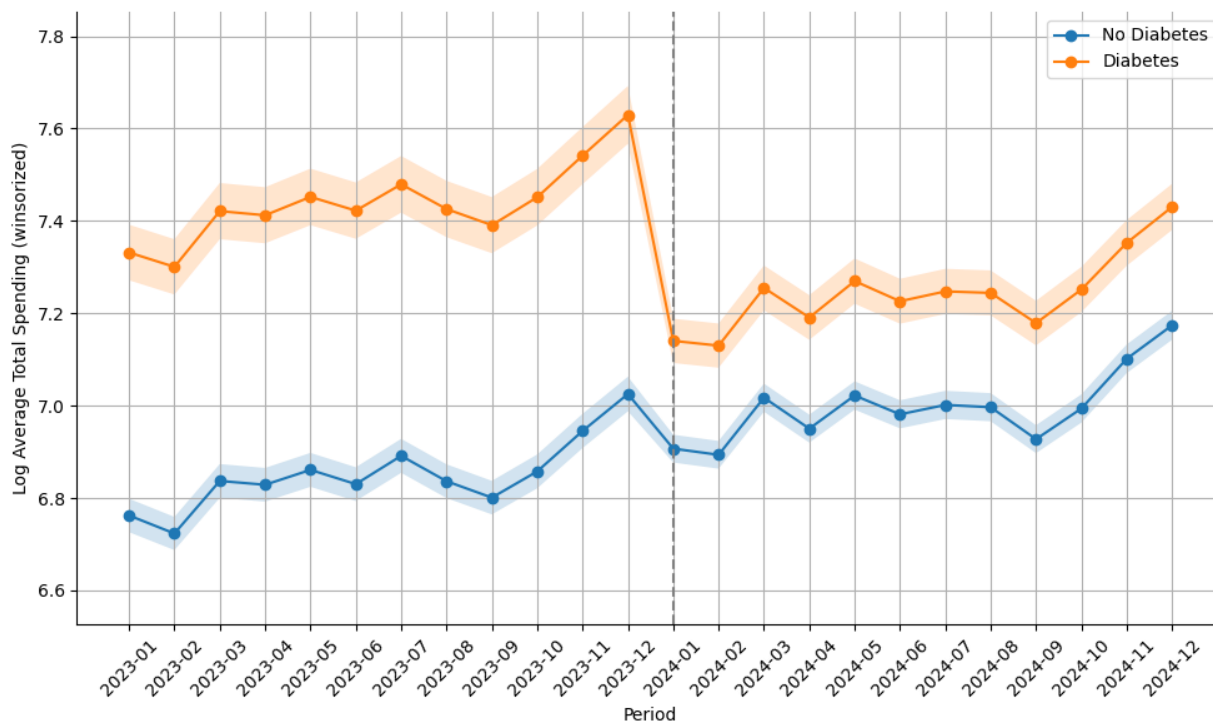
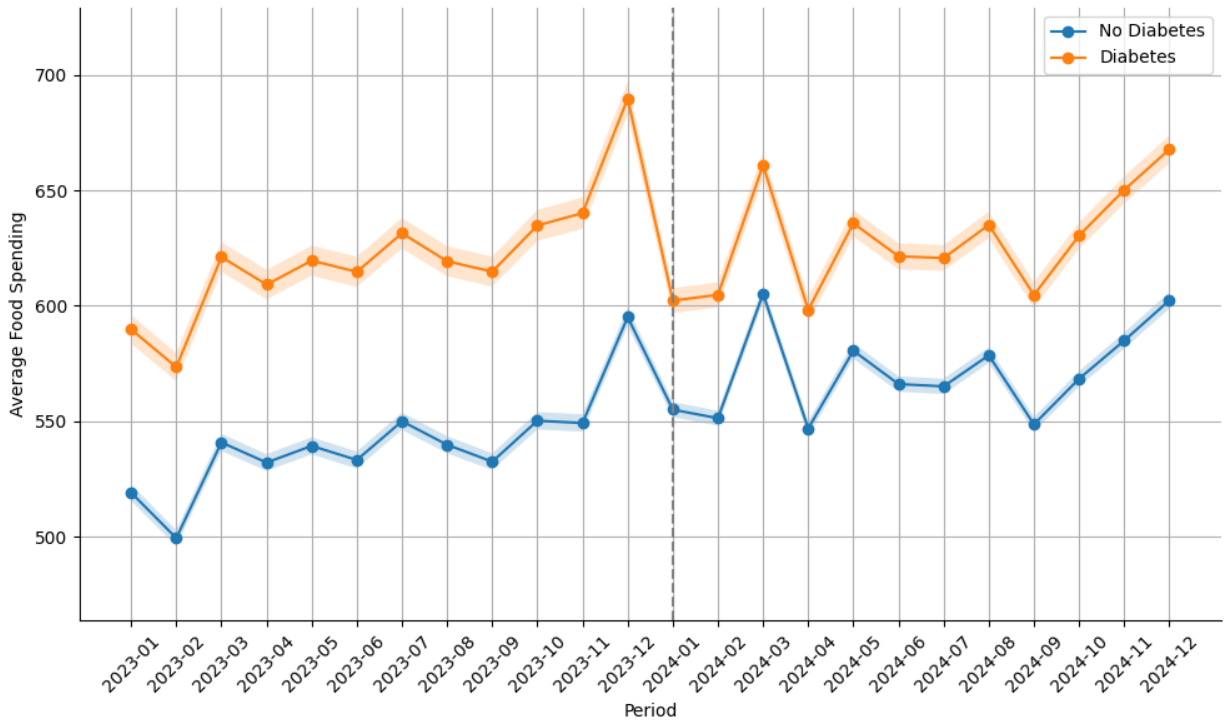


Figure 6: Monthly Average of Food Spending

(a) level



(b) log

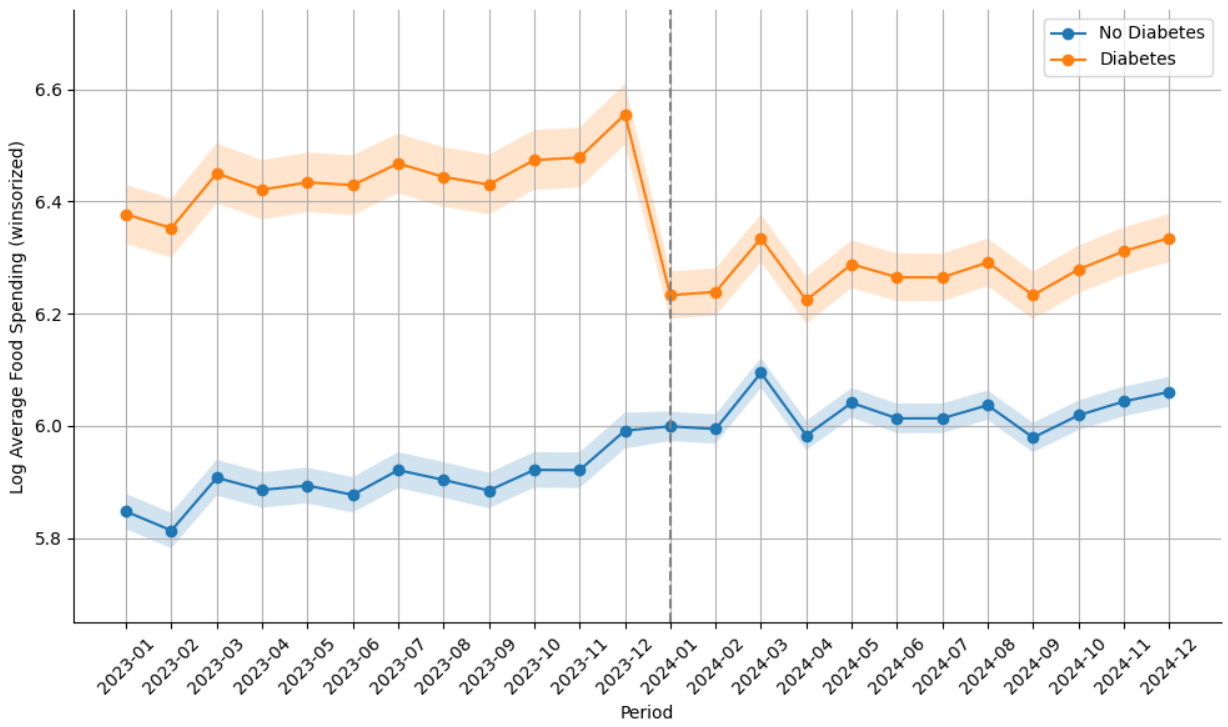
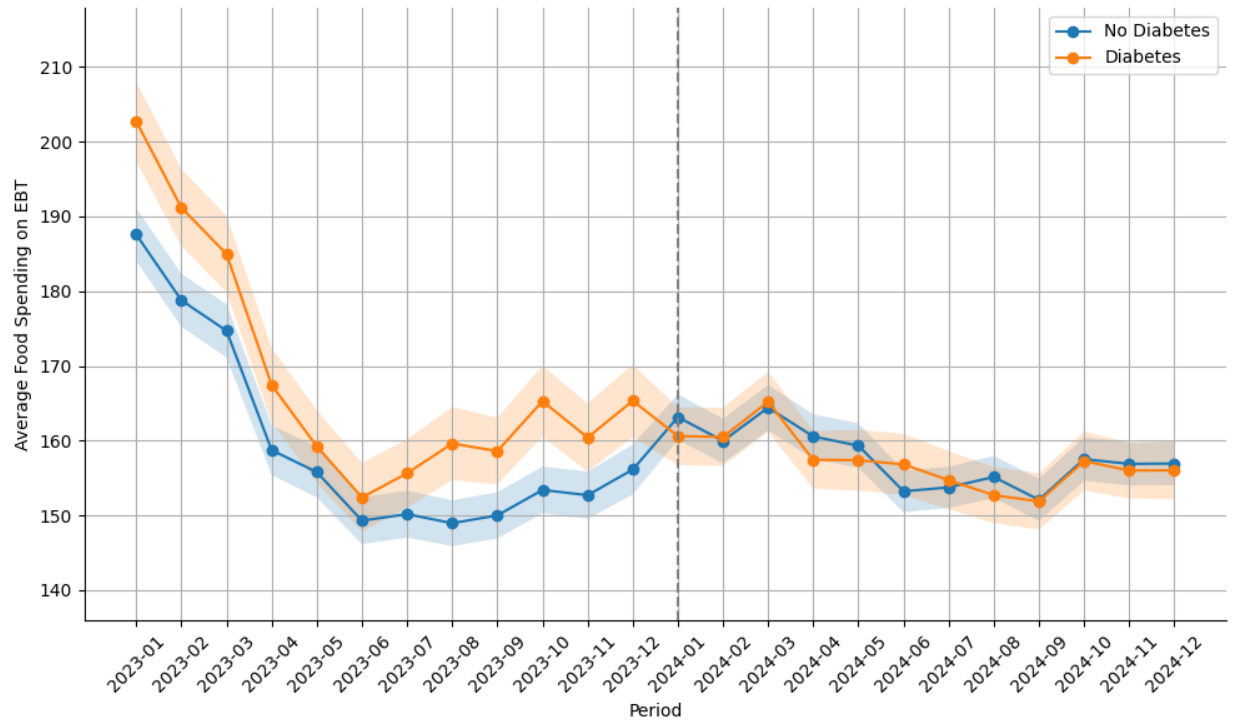


Figure 7: Monthly Average of Food Spending on EBT

(a) level



(b) log

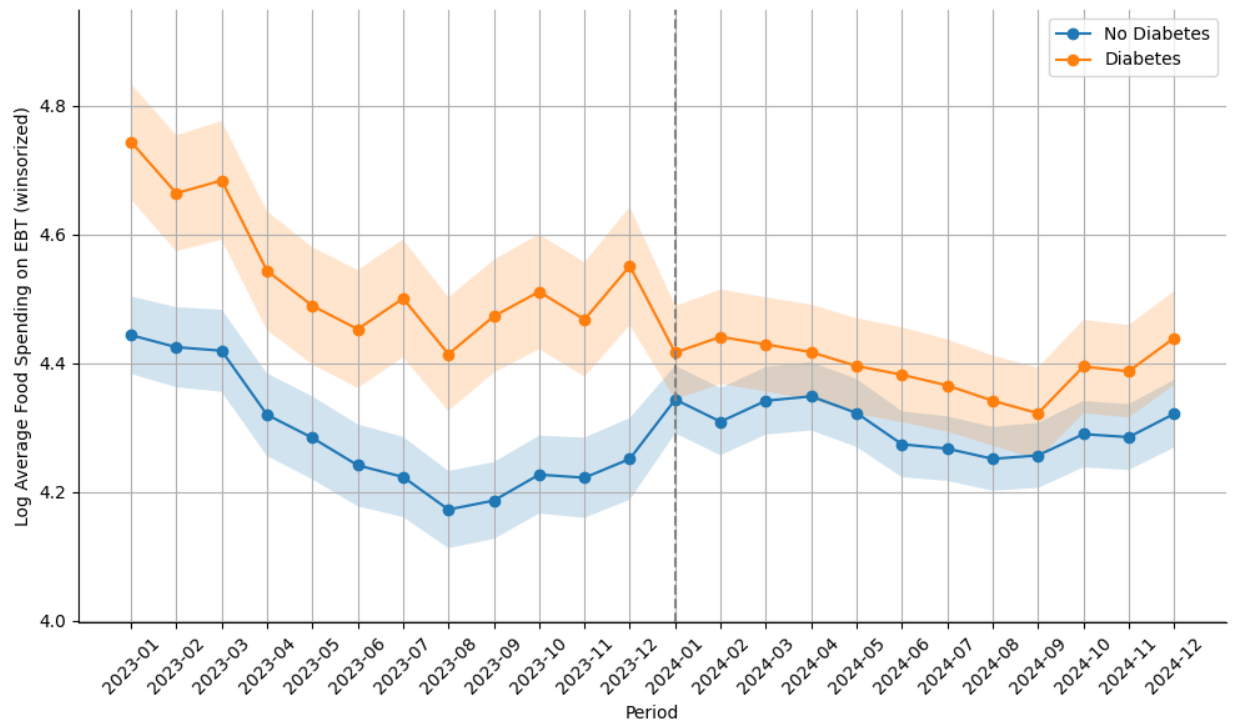
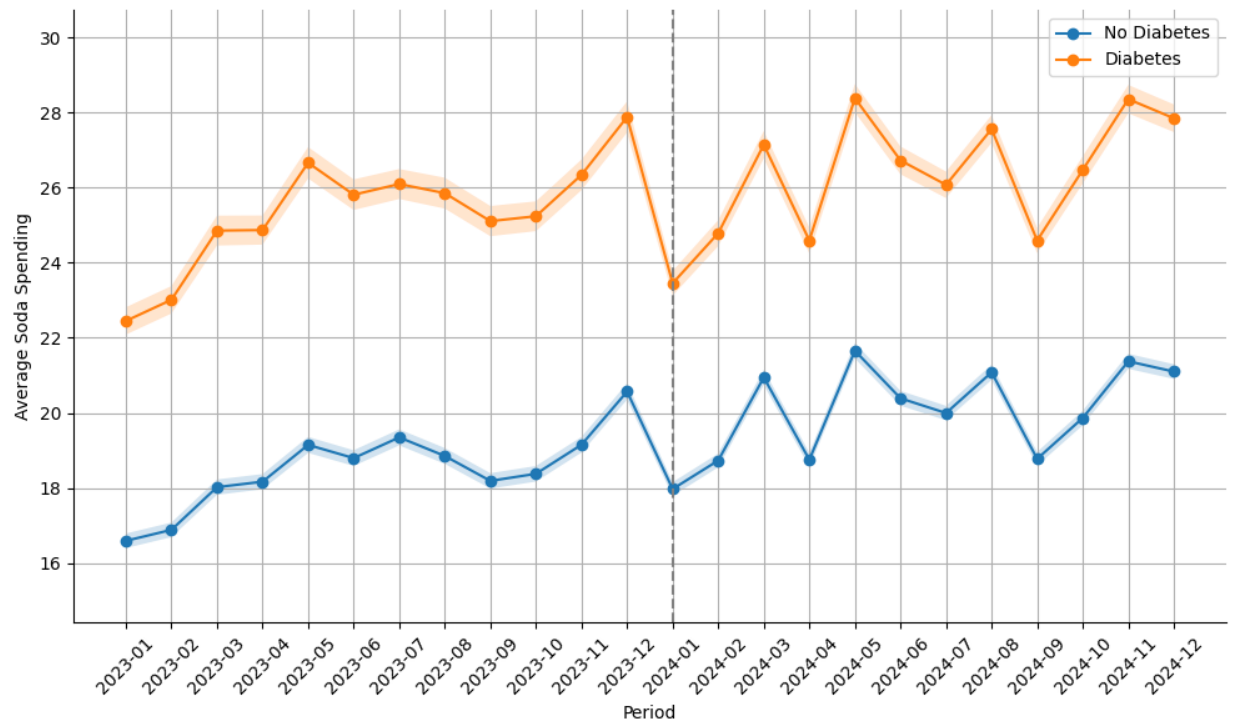


Figure 8: Monthly Average of Soda Spending

(a) level



(b) log

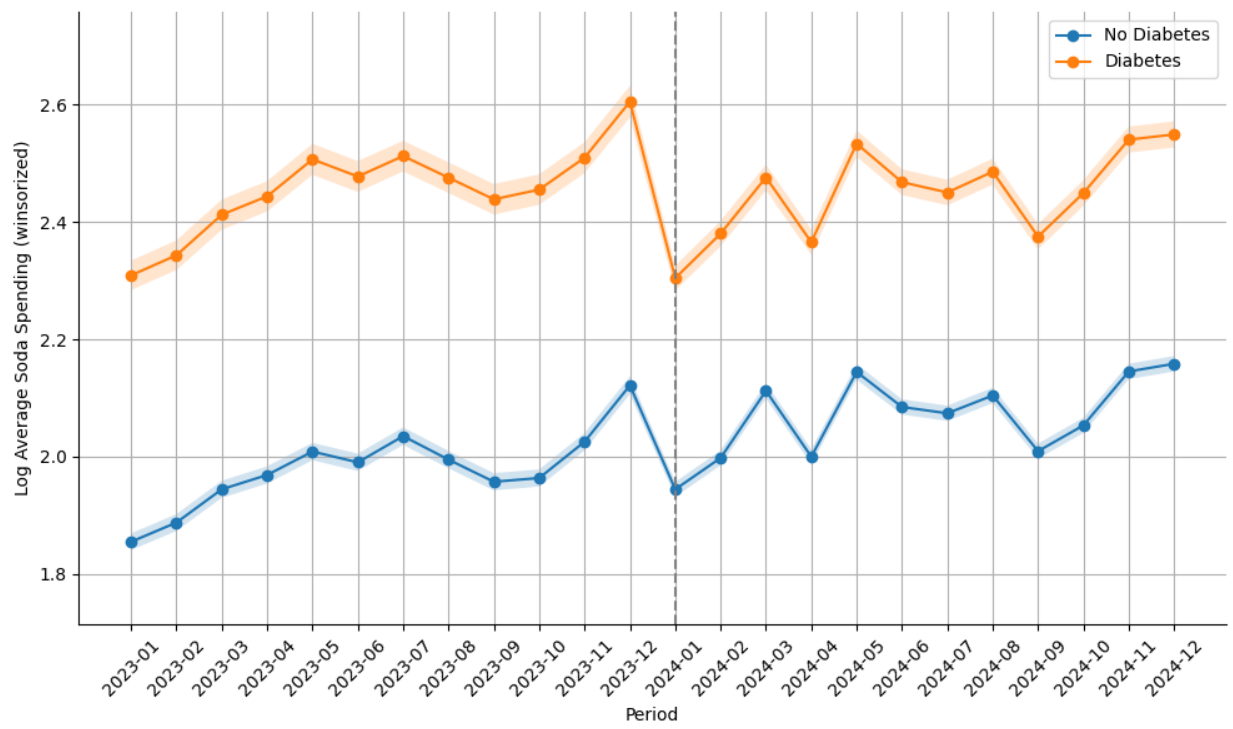
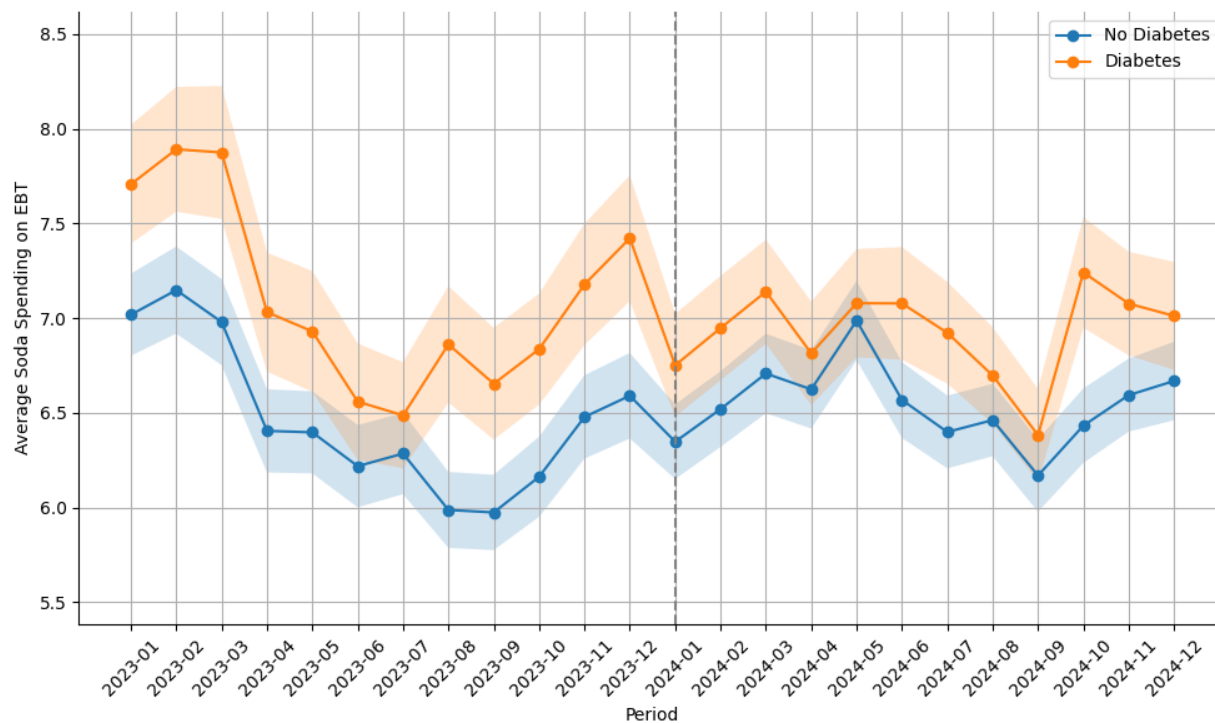


Figure 9: Monthly Average of Soda Spending on EBT

(a) level



(b) log

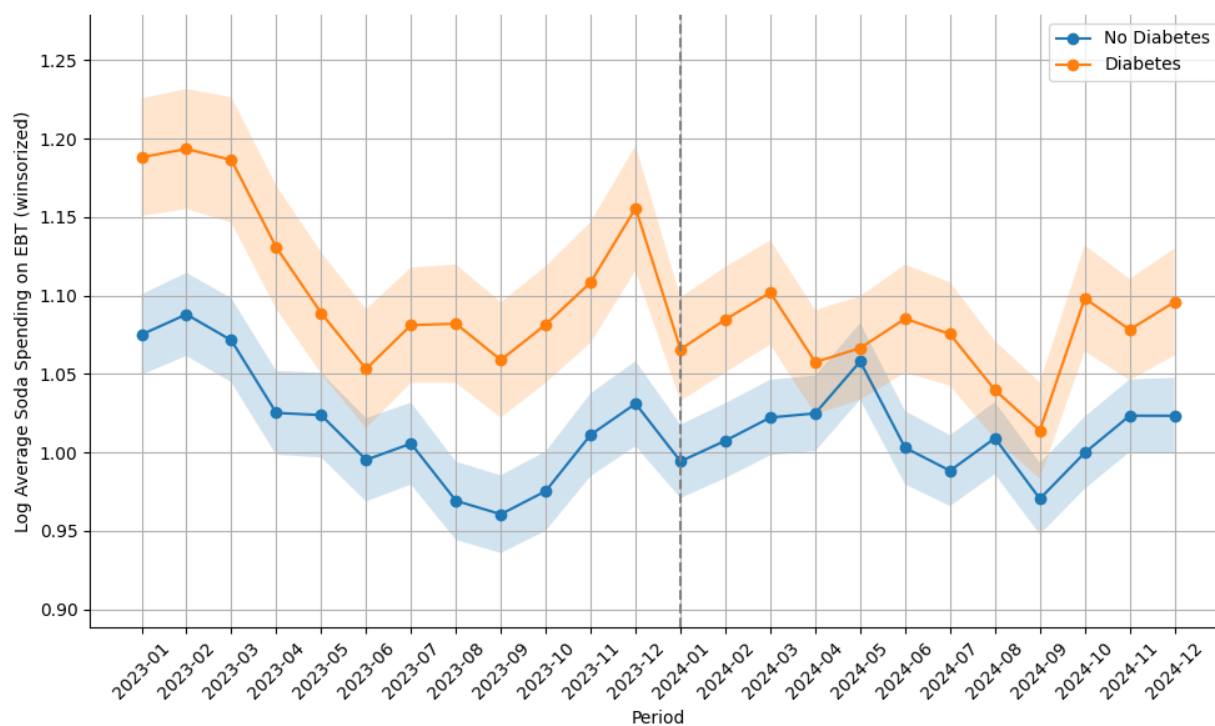


Figure 10: Proportion of Total Spending on Food

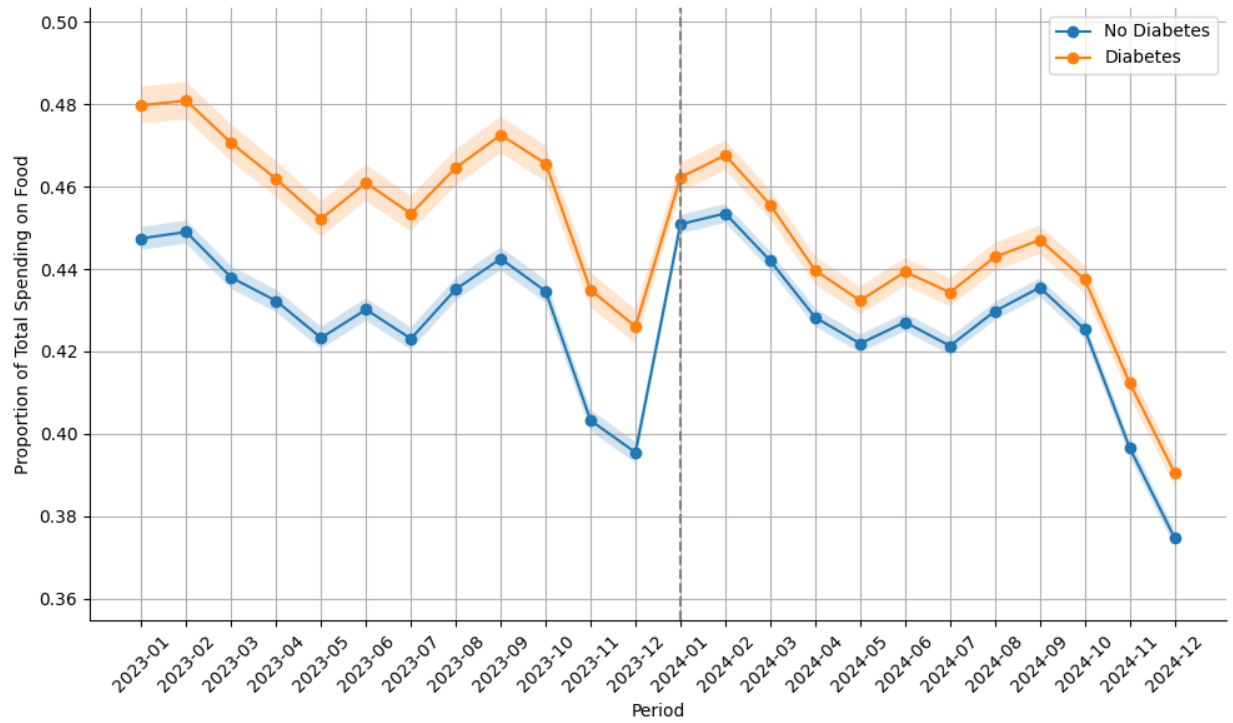


Figure 11: Proportion of Food Spending on EBT

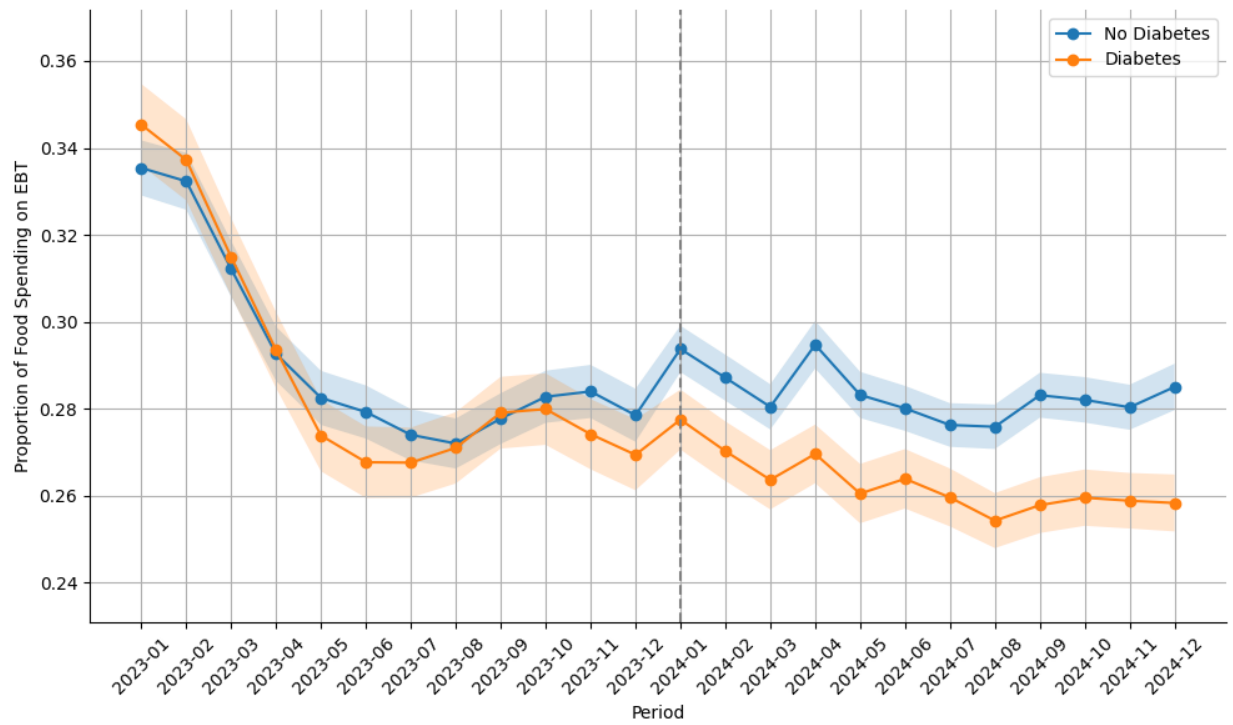


Figure 12: Proportion of Food Spending on Soda

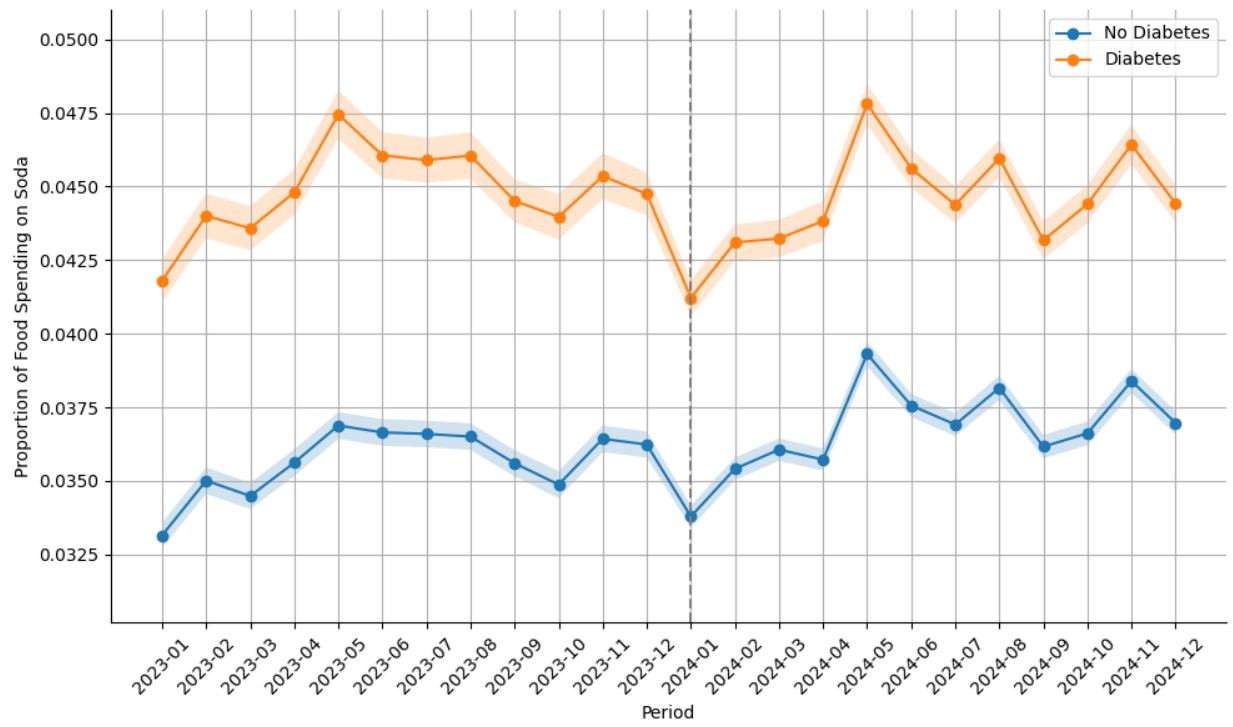


Figure 13: Proportion of EBT Spending on Soda

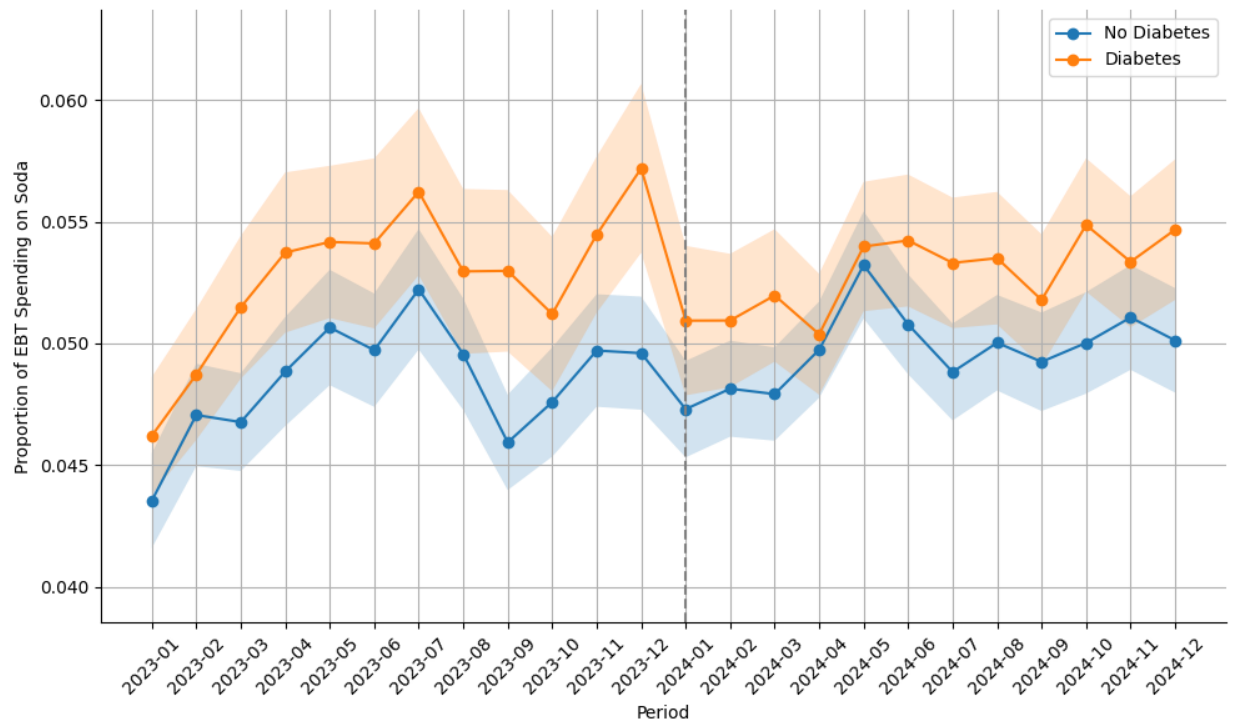


Figure 14: Proportion of Soda Spending on EBT

